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Super-resolution magnetotelluric data inversion with seismic texture constraint

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ABSTRACT

Magnetotelluric (MT) data inversion reconstructs the subsurface resistivity structure using measured natural electromagnetic (EM) fields. Due to the attenuation of low-frequency EM waves in geologic conductors, MT inversion has lower sensitivity and resolution for subsurface structures compared with seismic reflection imaging. Seismic reflection provides rich, detailed subsurface information that complements EM data, such as the spatial locations of geologic boundaries and fine-scale geophysical attributes. This work develops a super-resolution MT inversion approach incorporating the seismic texture constraint. We develop a multiresolution model parameterization scheme to represent the unknown model with the coarse-scale resistivity background and fine-scale resistivity details. A modified, pretrained VGG19 neural network (NN) serves as an effective texture extraction operator. This inversion method requires no NN training and simultaneously optimizes MT data misfit, texture discrepancy, and various regularization penalties using the adaptive moment estimation optimizer. Experiments show that our inversion can reduce or at least maintain the data misfit and effectively improve MT inversion resolution to a seismic-like level. This approach has promising application potential in the oil industry for highlighting quasi-layered subsurface structures.

INTRODUCTION

Magnetotelluric (MT) data inversion infers subsurface electrical structures using the natural electromagnetic (EM) field (Berdichevsky and Dmitriev, 2002). The MT method has been successfully applied in hydrocarbon exploration, analysis of deep earth structures, and hydrologic and geothermal exploration (Mansoori et al., 2016; Murphy and Egbert, 2017; Gao et al., 2018; Käufl et al., 2020; Ardid et al., 2021; Yang et al., 2021). MT inversion is sensitive to resistivity anomalies and is able to identify water-hydrocarbon boundaries (Hu et al., 2009). However, its spatial resolution is relatively low, making it challenging to accurately characterize stratigraphic interfaces (Berdichevsky et al., 1998; Ledo et al., 2002). In contrast, because of the higher resolution, seismic sections provide abundant geologic and geophysical information, such as large-scale features including strata interfaces and anomaly geometries (Yan et al., 2017), as well as finer-scale features, such as geophysical attribute textures (Gao, 2003). Integrating seismic information into MT inversion can improve reconstruction resolution and model consistency across various geophysical fields (Guo et al., 2020b; Zhou et al., 2022).

Studies suggest that seismic information can improve the delineation of structural boundaries in EM data inversion (Bedrosian, 2007; Moorkamp et al., 2016). In typical constrained inversions, seismic data or models guide gradient-based inversion through carefully designed regularizations (Bedrosian, 2007). Portniaguine and Zhdanov (1999) propose to modify the minimum-gradient support constraint by adjusting smoothness regularization weights based on wave velocity or seismic impedance models, improving the alignment of structural boundaries in EM and seismic models (Brown et al., 2012; Yan et al., 2017). Image-guided inversion approaches have also been developed to integrate high-resolution structural features into geophysical data inversion (Ma et al., 2010, 2012; Zhou et al., 2014a,

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2014b, 2016; Guo et al., 2017). Here, the non-Euclidean distances between pixels in the inverted EM model are calculated using seismic reflectivity, and these distances are used to adjust the smoothness regularization weights. In addition, Haber and Oldenburg (1997) introduce a structural penalty function to minimize the curvature discrepancies across various models, although the implicit requirement of this method may conflict with lithologic laws (i.e., simultaneous changes in different physical properties at the same location) (Bedrosian, 2007). As a remedy, the cross-gradient (CG) method (Gallardo and Meju, 2003), which highlights common boundaries where different models change simultaneously, has become widely used in joint and constrained inversions (Hu et al., 2009; Abubakar et al., 2012; Colombo et al., 2018). Recent studies have extracted resistivity boundaries from seismic reflection data using convolutional neural networks (CNNs) and incorporated them into EM data inversion (Oh et al., 2020; Zhou et al., 2023, 2024). Most existing research has focused on refining large-scale EM structures with seismic data, such as optimizing strata direction, position, and the geometry of anomalous regions. These structural boundaries typically align with strong reflection axes in seismic data. However, densely distributed weak reflection interfaces, which reveal textures of fine-scale stratum layers, have not been adequately used in existing MT (or EM) inversions.

Seismic texture analysis (STA) focuses on evaluating the distribution pattern of pixel amplitudes within a subregion of the seismic image (Love and Simaan, 2005). Gray-level cooccurrence matrices are used to represent the joint distribution of pixel values at different relative positions (Haralick et al., 1973), from which various statistics are derived to measure the texture (West et al., 2002; Gao, 2003; Chopra and Alexeev, 2006; Yenugu et al., 2010). STA is a valuable tool for characterizing and predicting reservoir parameters, providing insights into the distribution, volume, and connectivity of hydrocarbon-bearing facies (Chopra and Alexeev, 2006; Yenugu et al., 2010). In geophysics, as different geophysical parameters are correlated (Mukerji et al., 2001; Grana and Della Rossa, 2010), seismic texture also reflects the fine distribution of resistivity, which is generally difficult to be depicted due to the low-resolution and limited sensitivity of MT data inversion.

Image fusion techniques have been widely used across various imaging applications, such as remote sensing (Pandit and Bhiwani, 2015), biomedicine and medical diagnosis (Cui et al., 2023), industrial defect detection (Hu et al., 2021), and archaeological site analysis (Karamitrou et al., 2020). These techniques typically synthesize data from multiple sensing methods (e.g., optical cameras, millimeter wave cameras, infrared, and near-infrared imaging), each capturing unique aspects of the subject. By combining these multidimensional data sets, researchers create a single comprehensive image that preserves critical information and surpasses the quality of individual input sources (Omar and Stathaki, 2014). Recently, Wei et al. (2024) extend this approach to geophysical research by proposing a fusion method that integrates geophysical magnetic data with geochemical data sets. Their work marks one of the earliest efforts to integrate geophysical and geochemical data in natural resource exploration.

As a typical image fusion task in computer vision (CV) research, image style transfer allows rendering a content image in the style of another with machine-learning and nonmachine-learning approaches (Efros and Leung, 1999; Wei and Levoy, 2000). Since the 2010s, deep-learning (DL) techniques have been widely applied to image style transfer with notable advancements achieved (Gatys et al., 2016; Huang and Belongie, 2017; Zhu et al., 2017; Kim et al., 2019). CNN-based image feature extractors trained on general image data sets (e.g., ImageNet) have been extensively studied, with feature map statistics at different levels proven to be effective style representations (Gatys et al., 2016; Huang and Belongie, 2017). In addition, these operators, referred to here as image texture extractors, are differentiable, which allows them to be seamlessly integrated as a penalty term in gradient-based EM inversion.

In this work, we study a super-resolution MT data inversion incorporating a seismic texture constraint. The texture extractor is based on a VGG19 neural network (Simonyan, 2014), pretrained on the ImageNet data set (Deng et al., 2009), to define texture measurements for geophysical images. In the inversion, we simultaneously minimize the MT data misfit, the texture discrepancy between the seismic section and the inverted model, and additional regularization terms using the adaptive moment estimation (ADAM) method (Kingma and Ba, 2017). Our results show that the inverted resistivity model captures fine-scale textures of geophysical attribute distribution, achieving a lower or comparable final data misfit to conventional gradient-based inversion. This approach has promising practical value, particularly for highlighting quasi-layered subsurface structures.

METHODS

MT forward problem

In 2D MT forward modeling, Maxwell's equations can be decoupled into the transverse magnetic (TM) mode and the transverse electric (TE) mode. Assuming the subsurface structure extends along the x-direction and denoting the time harmonic term as $e^{j\omega t}$, the TM mode equation can be written as

$$\frac{\partial}{\partial y}\left(\frac{1}{\sigma+j\omega\epsilon}\frac{\partial H_x}{\partial y}\right)+\frac{\partial}{\partial z}\left(\frac{1}{\sigma+j\omega\epsilon}\frac{\partial H_x}{\partial z}\right)-j\omega\mu H_x=0, \quad (1)$$

with the boundary condition

$$\frac{\partial H_x}{\partial y}|_{\mathrm{PQ,MN}}=0, H_x|_{\mathrm{QN}}=0, H_x|_{\mathrm{PM}}=1, \quad (2)$$

whereas the TE mode equation can be written as

$$\frac{\partial}{\partial y}\left(\frac{1}{j\omega\mu}\frac{\partial E_x}{\partial y}\right)+\frac{\partial}{\partial z}\left(\frac{1}{j\omega\mu}\frac{\partial E_x}{\partial z}\right)-(\sigma+j\omega\epsilon)E_x=0, \quad (3)$$

with the boundary condition

$$\left.\frac{\partial E_x}{\partial y}\right|_{\mathrm{PQ,MN}}=0, \quad E_x|_{\mathrm{QN}}=0, \quad E_x|_{\mathrm{PM}}=1. \quad (4)$$

Here, PQ, MN, QN, and PM denote the boundaries of the computational domain, as shown in Figure 1. The variables E_x and H_x represent the x-component electric and magnetic fields, respectively. The parameters σ, ϵ, and μ denote electrical conductivity, electric permittivity, and magnetic permeability, respectively, and ω denotes the angular frequency. We use the finite-difference method (FDM) to numerically compute the subsurface EM field. Further information about the forward solver can be found in Guo et al. (2020a).

In the MT system, the surface electric and magnetic field components (E^0 and H^0) are measured at receivers (indicated by red triangles in Figure 1), and apparent resistivity and phase are calculated with

$$\rho_{\mathrm{TM}} = \frac{1}{\omega\mu}\left|\frac{E_y^0}{H_x^0}\right|^2, \quad \phi_{\mathrm{TM}} = \arctan\left(\frac{E_y^0}{H_x^0}\right), \tag{5}$$

for TM mode and with

$$\rho_{\mathrm{TE}} = \frac{1}{\omega\mu}\left|\frac{E_x^0}{H_y^0}\right|^2, \quad \phi_{\mathrm{TE}} = \arctan\left(\frac{E_x^0}{H_y^0}\right), \tag{6}$$

for TE mode.

In this work, the MT forward modeling and related operations are implemented using TensorFlow (Abadi et al., 2016), an open-source machine-learning framework developed by Google that facilitates the building, training, and deployment of DL models across various applications. Rather than training a surrogate model on a large data set, we leverage DL techniques to encapsulate MT forward modeling within an NN interface. Specifically, we implement the FDM-based forward problem in TensorFlow using the *tensor* data format, providing a key advantage: in the inversion, automatic differentiation (Baydin et al., 2018) can be used to compute Jacobian matrices. In layered model simulations, the forward responses computed by our method are within a 0.5% relative error threshold compared with the analytical solution. Using a mesh size of approximately tens by tens, each forward simulation requires approximately 1 s on an Nvidia Tesla V100 graphics processing unit (GPU).

Texture extraction operator

Over the past decades, CNNs have become an important tool in computational science. CNNs typically consist of cascaded convolutional layers, enabling the extraction of hierarchical features from input data. In our approach, we design the texture extraction operator with two cascaded components: a preprocessing part and a statistical feature extraction part.

1) Image preprocessing: The input model $\mathbf{m}$ is processed with peak clipping at the thresholds $\pm K$ (K is a positive real number) and then normalized to the 0–1 range. The formulas are

$$m_{p,x,y} = \begin{cases} -K, & \text{if } m_{x,y} \le -K, \ \forall\, x,y \\ m_{x,y}, & \text{if } -K < m_{x,y} < K, \ \forall\, x,y \\ K, & \text{if } m_{x,y} \ge K, \ \forall\, x,y \end{cases} \tag{7}$$

and

$$\mathbf{m}_n = \frac{\mathbf{m}_p - \min\{\mathbf{m}_p\}}{\max\{\mathbf{m}_p\} - \min\{\mathbf{m}_p\}}, \tag{8}$$

where x and y represent the pixel coordinates within the model. Here, $\mathbf{m}$, $\mathbf{m}_p$, and $\mathbf{m}_n$ are model vectors and $m_{p,x,y}$ and $m_{x,y}$ are components of $\mathbf{m}_p$ and $\mathbf{m}$ at the coordinate (x, y).

2) Statistical feature extraction: The preprocessed image $\mathbf{m}_n$ is fed into a CNN, and we use $G_p^q(\mathbf{m}_n)$ to denote the 2D feature map generated by the qth output channel at the pth convolutional layer. The total number of elements in $G_p^q(\mathbf{m}_n)$ is denoted with N_p^q. The mean and variance of $G_p^q(\mathbf{m}_n)$ are calculated as follows:

$$\mu_p^q = \frac{1}{N_p^q}\sum_{i=1}^{N_p^q} G_{p,i}^n(\mathbf{m}_n), \quad \sigma_p^{2q} = \frac{1}{N_p^q}\sum_{i=1}^{N_p^q}(G_{p,i}^q(\mathbf{m}_n) - \mu_p^q)^2. \tag{9}$$

The texture measurement $\mathbf{s}$ is a weighted combination of means and variances from various convolutional layers and channels as

$$\mathbf{s} = \{\omega_p^q \mu_p^q, \nu_p^q \sigma_p^{2q}, p = 1, 2\ldots M, q = 1, 2\ldots N_p\}, \tag{10}$$

where $\{.\}$ denotes the set of variables. Here, M and p denote the total number of convolutional layers and the pth convolutional layer among them, and N_p and q denote the total number of the output channels of the pth convolutional layer and the qth output channel among them. The coefficients ω_p^q and ν_p^q adjust the weights of different components. The texture data $\{\mu, \sigma^2\}_p^{\tilde{q}}$ ($\tilde{q}$ represents any q) from various convolutional layer p provide texture characterization at different abstraction levels (Gatys et al., 2016). The weights $\{\omega, \nu\}_p^{\tilde{q}}$ can be flexibly adjusted based on experience or preferences. In our study, experiments indicate that the means and variances from a single convolutional layer (e.g., the 12th convolutional layer) sufficiently form the texture measurement.

Letting $\mathbf{m}$ represent the input image, we denote the cascaded operations introduced in the preceding as $\mathcal{T}$, thereby expressing texture extraction from an image with

$$\mathbf{s} = \mathcal{T}(\mathbf{m}). \tag{11}$$

In this work, specifically, $\mathcal{T}$ is implemented using the VGG19 neural network pretrained on the ImageNet data set, which comprises more than 1.2 million real-world images across more than 1000 categories (Deng et al., 2009; Simonyan, 2014). We collect the means and variances of the output feature maps from the 12th

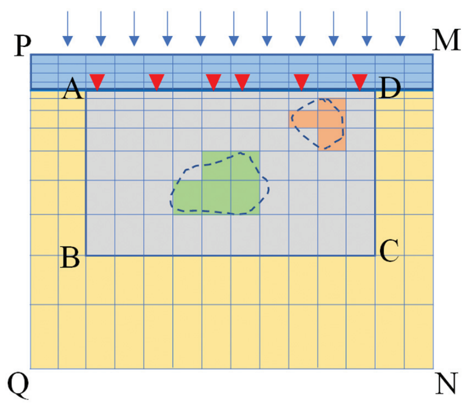

Figure 1. A diagram of the mesh discretization for 2D MT forward modeling. The red inverted triangles represent the MT receivers on the earth surface, and the blue arrows denote the incident plane EM wave. The upper blue region, gray region, and yellow region at the flank and bottom, respectively, denote the air, DoI, and the extension of the DoI to guarantee the accuracy of forward modeling. The green and orange regions within the DoI represent the geophysical anomalies.

convolutional layer (as shown in Figure 2), with equal weights assigned to these statistics across all output channels. Specifically, the coefficients for the means and variances are set as follows:

$$\omega_p^q, \nu_p^q = \begin{cases} 1, & \text{if } p = 12, \ \forall\, q \\ 0, & \text{if } p \neq 12, \ \forall\, q \end{cases}. \tag{12}$$

Hence, the texture measurement $\mathbf{s}$ is given by

$$\mathbf{s} = \{\mu_{12}^q, \sigma_{12}^{2q}, q = 1, 2 \ldots 512\}. \tag{13}$$

The image height H and width W can be set to any integer multiple of eight, so the resulting texture measurement is a fixed vector of 1024 elements, independent of the input image size.

Model parameterization scheme

In the proposed method, a dual parameterization scheme is used. The unknown resistivity model is parameterized by two variables $\mathbf{m}_1$ and $\mathbf{m}_2$. Here, $\mathbf{m}_1$ represents the low-resolution base resistivity model on the coarse-scale EM grid (typically tens by tens in size), and $\mathbf{m}_2$ is the high-resolution detail resistivity model on the fine-scale seismic grid (typically hundreds by hundreds). The interpolation function $\mathcal{I}_{c_1}$ maps a model from the coarse-scale EM grid to the fine-scale seismic grid, and $\mathcal{I}_{c_2}$ maps a model from the seismic grid to the EM grid. Consequently, the total resistivity models on both grids are the sum of base and detail models on the same grid, that is,

$$\mathbf{m}_{\text{total}}^{\text{coarse}} = \mathbf{m}_1 + \mathcal{I}_{c_2}(\mathbf{m}_2), \tag{14}$$

$$\mathbf{m}_{\text{total}}^{\text{fine}} = \mathcal{I}_{c_1}(\mathbf{m}_1) + \mathbf{m}_2, \tag{15}$$

where $\mathbf{m}_{\text{total}}^{\text{coarse}}$ and $\mathbf{m}_{\text{total}}^{\text{fine}}$ represent the resistivity of the same computational area but at different resolutions: the former at EM resolution and the latter at seismic resolution. The MT response is calculated based on the coarse-scale total model $\mathbf{m}_{\text{total}}^{\text{coarse}}$, and the texture measurement is calculated based on the fine-scale detail model $\mathbf{m}_2$.

In this work, whereas $\mathcal{I}_{c_1}$ is implemented using rectangular bivariate spline (RBS) interpolation (Chui, 1988), $\mathcal{I}_{c_2}$ is constructed using an area-weighted resampling scheme (Jin et al., 2018) (i.e., determining the relative weights of all pixels in the fine model based on the area they overlap with corresponding pixels in the coarse model). We

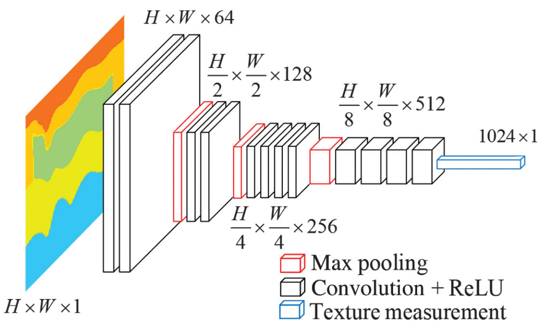

Figure 2. Diagram of the texture extractor. Here, ReLU represents rectified linear unit activation function.

implement $\mathcal{I}_{c_2}$ with the area-weighted resampling scheme instead of RBS for two main reasons. First, because of the inherent locality, using RBS interpolation can leave many fine model parameters underconstrained. Second, because area-weighted resampling is a linear transform, it is easy to compute the Jacobian matrix of $\mathcal{I}_{c_2}$ with the area-weighted scheme compared with its nonlinear counterpart (RBS interpolation).

Inversion constrained by seismic texture

Finally, the objective function for the proposed inversion is formulated as

$$\begin{aligned}\mathcal{L}(\mathbf{m}_1, \mathbf{m}_2) = {} & \alpha_D \|\mathbf{d}_{\text{obs}} - \mathcal{F}(\mathbf{m}_1 + \mathcal{I}_{c_2}(\mathbf{m}_2))\|^2 \\ & + \lambda_L \|\nabla(\mathbf{m}_1 + \mathcal{I}_{c_2}(\mathbf{m}_2))\|^2 \\ & + \lambda_S \|\mathcal{T}(\mathbf{m}_{\text{seis}}) - \mathcal{T}(\mathbf{m}_2)\|^2 + \lambda_M \|\mathbf{m}_2\|^2,\end{aligned} \tag{16}$$

where α_D is the normalization hyperparameter and λ_L, λ_S, and λ_M are hyperparameters adjusting the relative weights of different terms. Here, $\mathbf{d}_{\text{obs}}$ and $\mathbf{m}_{\text{seis}}$ represent the measured MT data and the seismic reflection section, respectively. The term ∇ denotes the gradient operator, and F, T, and $\mathcal{I}_{c_2}$ refer to the MT forward problem, texture extraction operator, and area-weighted resampling function, respectively, as introduced in the subsections "MT forward problem," "Texture extraction operator," and "Model parameterization scheme." The unknowns in equation 16 are $\mathbf{m}_1$ and $\mathbf{m}_2$, corresponding to the base and detail resistivity models introduced in the subsection "Model parameterization scheme." When the objective function (equation 16) converges, the final fine-scale model can be calculated with equation 15, which serves as the expected final output of the proposed inversion. As long as all forward computations included in equation 16 use TensorFlow tensors, equation 16 can be readily optimized on a GPU using ADAM optimizer in TensorFlow. The loss function diagram for the proposed inversion is shown in Figure 3. The code of this work is available in the GitHub repository.

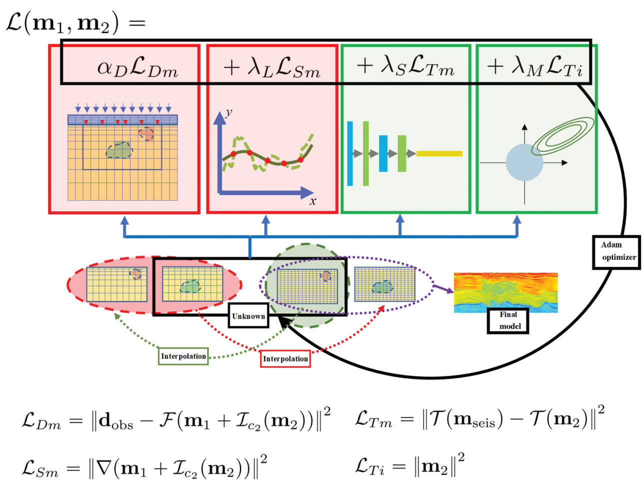

Figure 3. Diagram of the proposed MT inversion constrained with seismic texture. The abbreviations Dm, Sm, Tm, and Ti denote data misfit, smoothness, texture misfit, and texture intensity, respectively.

NUMERICAL EXPERIMENTS

Feasibility study

Because the adopted NN (inspired by the AdaIN layer in Huang and Belongie [2017]) is originally designed for image style transfer, it is necessary to test the proposed method's capability to incorporate texture before conducting inversion experiments. Earlier works have attempted to define image style using similarity measures or local statistics based on pixel values. A widely accepted definition describes the image style as the visual representation determined by color distribution, texture, stroke, brightness, contrast, and so on (Prokhorov, 1983). Notably, the style constraint is broader than the texture constraint. Style consistency generally does not enforce the spatial correspondence between images. However, because different geophysical properties of the same structure are intrinsically coupled, directly applying seismic style constraints to EM inversion may lead to physically unrealistic artifacts.

We verify through numerical experiments that the designed texture extraction operator enforces a texture consistency constraint rather than merely a style one. Our goal is to recover the target image (shown in Figure 4a, a simulated 224×224 seismic section of the Marmousi model [Brougois et al., 1990]) using gradient-based local optimization (e.g., the ADAM method) from a blank initial image. The baseline experiment (see Figure 4b) demonstrates that with the texture extraction operator configured as in Huang and Belongie (2017), the reconstructed image lacks sufficient spatial correspondence with the target. For example, in the middle-right region with depth ranging between 0.8 and 2.2 km, the texture orientation of the reconstructed image deviates from the target image, and abnormal uneven white spots also appear. To address these challenges, we divide the target and reconstructed images into $n \times n$ patches and minimize the summation of texture misfits between the spatially corresponding patches in both images. Dividing the $W \times H$ image into $n \times n$ patches involves partitioning the width and height into n segments, yielding n^2 patches, each with dimensions $(W/n) \times (H/n)$. Here, n is set to two, three, and five, respectively. With finer-grained patches, the spatial texture alignment between the target and reconstructed images improves significantly (see Figure 4c–4e). In addition, slight discontinuities may appear where different patches intersect; however, these can be mitigated in constrained inversion by carefully setting the initial detail model and regularization penalties.

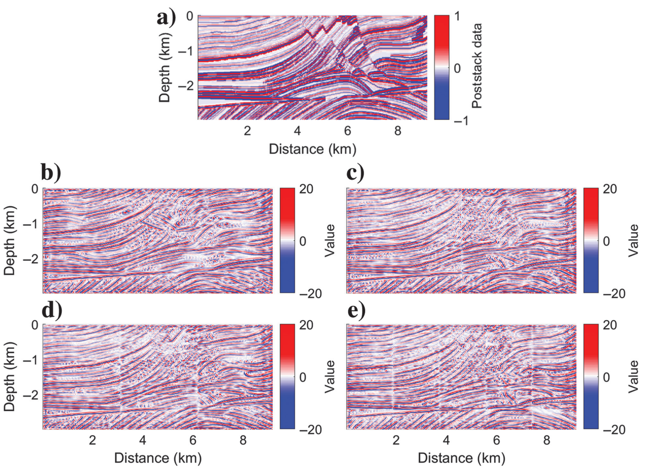

Figure 4. (a) Target image and reconstructed images with different division approaches: (b) no division applied, (c) 2×2 division, (d) 3×3 division, and (e) 5×5 division.

Inversion of Marmousi model

The Marmousi model is widely used as a test benchmark for seismic imaging and reverse time migration algorithms because its complex structure offers extensive opportunities for validating these algorithms. In this study, we use the original Marmousi model. The domain of investigation (DoI) spans 9.2 km horizontally and 3 km vertically. Synthetic resistivity values are generated by mapping the Marmousi velocity model to a resistivity range of 10 to 100 Ωm. Assuming a uniform subsurface density, the reflection coefficients are calculated from the synthetic velocity using

$$R = \frac{v_2 - v_1}{v_2 + v_1}, \tag{17}$$

where v_1 and v_2 are the velocities above and below an interface. This assumption is made because we use only the geometric structure of the Marmousi model. As shown subsequently, the proposed method does not rely on any a priori knowledge of lithologic relationships, so it is unnecessary to distinguish whether changes in acoustic impedance arise from variations in density or velocity. The reflection coefficients are then convolved with an 8 Hz Ricker wavelet to simulate the depth-domain seismic section. The seismic section is preprocessed as introduced previously with the peak clipping threshold K set to 1. The EM and seismic grids used in the experiment are 64×50 and 224×224, respectively. The fine synthetic model is mapped to the EM grid to generate simulated MT data. The 2D TE mode forward modeling and 22 working frequencies ranging from $10^{-2.5}$ to $10^{3.5}$ Hz are used, and 5% Gaussian noise is added to the simulated apparent resistivity and impedance phase. The synthetic resistivity model (on the fine grid, which is also the true model), simulated seismic section, apparent resistivity, and impedance phase are shown in Figure 5.

The detail model $\mathbf{m}_2$ and the seismic section $\mathbf{m}_{\text{seis}}$ are uniformly divided into 25 patches as mentioned in the subsection "Feasibility study." We respectively conduct two different inversions, which use the same inversion framework as elaborated in the "Method" section but differ in two aspects. First, the initial model for the first inversion is generated by interpolating the measured apparent resistivity (Figure 5c) to the DoI (as shown in Figure 6a), whereas the initial model for the second inversion (Figure 6b) is the sum of the

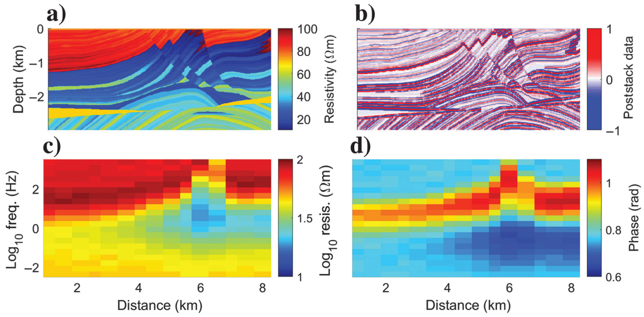

Figure 5. (a) Synthetic resistivity model, (b) seismic section after peak clipping, (c) apparent resistivity, and (d) impedance phase.

interpolated apparent resistivity and the seismic section multiplied by a proper coefficient (i.e., the base and detail parts, as shown in Figure 6c–6d). Second, the texture misfit hyperparameters (λ_S) for the first and second inversions are zero and nonzero, respectively. For simplicity, in the following discussions, the two inversions are referred to as conventional inversion and constrained inversion, respectively, and the constrained inversion discussed with the specified hyperparameters is referred to as baseline inversion.

The inverted models of the conventional inversion and baseline inversion as well as the base and detail models of baseline inversion are shown in Figure 7a–7d. In Figure 8, we also plot traces of 3, 5, and 7 km offset from the synthetic Marmousi model (Figure 5a), initial models (Figure 6a and 6b), and final models (Figure 7a and 7b). The hyperparameters λ_S, λ_M, and λ_L for baseline inversion are 1×10^{-8}, 2×10^{-8}, and 1×10^{-3}, respectively. The conventional and constrained inversion converge in 160 steps with final data misfit being 0.59% and 0.62%, respectively. The convergence curves are shown in Figure 9, and the predicted MT data and pixel-wise relative errors are shown in Figure 10. We run the experiments on a server with one Intel Xeon Platinum 8255C @ 2.5 GHz CPU and one Nvidia Tesla V100 GPU, and each inversion for the synthetic experiment takes approximately 1.5 h. The data misfit DM and the pixel-wise relative error $\mathbf{r}$ can be calculated with the observed data $\mathbf{d}_{\mathrm{obs}}$ and the predicted data $\mathbf{d}_{\mathrm{pre}}$ using

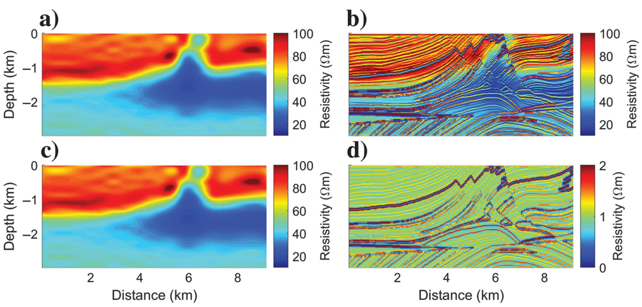

Figure 6. (a) Initial model for conventional inversion, (b) initial total model for constrained inversion, and (c and d) base and detail parts of the initial model for constrained inversion.

$$\mathrm{DM}=\frac{\|\mathbf{d}_{\mathrm{pre}}-\mathbf{d}_{\mathrm{obs}}\|}{\|\mathbf{d}_{\mathrm{obs}}\|}\times100\% \tag{18}$$

and

$$\mathbf{r}=\frac{\mathbf{d}_{\mathrm{pre}}-\mathbf{d}_{\mathrm{obs}}}{\mathbf{d}_{\mathrm{obs}}}. \tag{19}$$

We also compute the inverted models on the coarse grid with equation 14, as shown in Figure 7e–7h. The two final models on the coarse grid are similar, although Figure 7f seems less smooth than Figure 7e. The normalized L_2-norm model discrepancies between Figure 7f and 7g and between Figure 7f and 7h are 7.22% and 102.05%, respectively. These suggest that although the fine grid

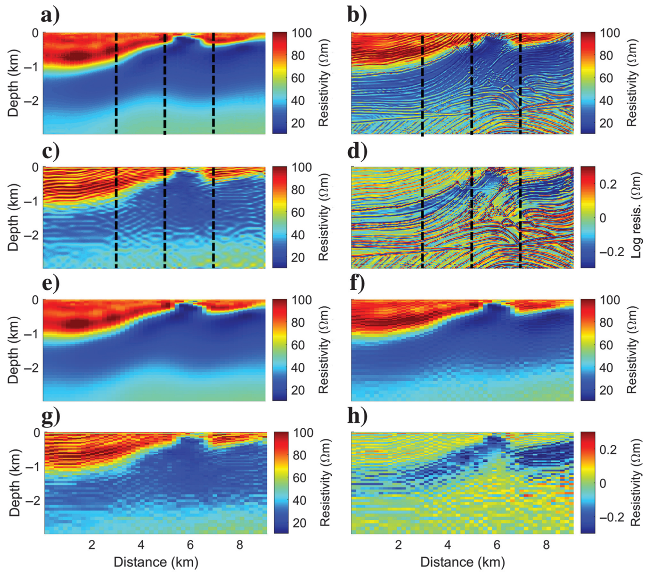

Figure 7. (a) The final model of the conventional inversion on the fine grid, and (b–d) the final total, base, and detail models of the constrained inversion on the fine grid. Here, (e–h) correspond to models (a–d) interpolated to the coarse grid. The dotted black lines in (a–d) represent the positions of traces in Figure 8.

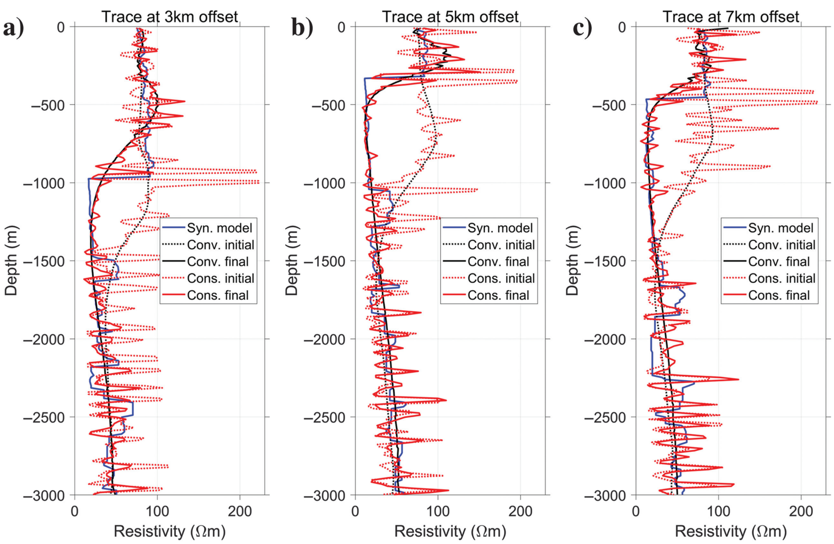

Figure 8. Comparison of the synthetic Marmousi model and corresponding initial and reconstructed traces at (a) 3 km offset, (b) 5 km offset, and (c) 7 km offset. The abbreviations Syn., Conv. and Cons. denote synthetic, conventional, and constrained, respectively.

model $\mathbf{m}_2$ introduces detailed structures beyond the MT resolution to the final model, the coarse-scale final model remains similar to that obtained with the conventional inversion. Thus, the failure risk of the MT forward operator due to the detailed $\mathbf{m}_2$ during the inversion is largely mitigated, providing strong support for the credibility of the proposed inversion. The data misfit of the forward response for Figure 7g is 2.25%, which is larger than in Figure 7f, highlighting the added value of the texture in fitting the MT data.

In conventional inversion, the boundary between the near-surface high-resistivity layer and the mid-depth conductive layer can be accurately located, and the structure of approximately 50 Ωm beneath the conductive layer can be depicted in the inverted model. However, the inverted model appears blurred compared with the synthetic model and has limited ability to depict resistivity variations, especially in deeper structures. With the constrained inversion, the stratum interfaces are more sharply defined (see the solid blue and red lines in Figure 8). The texture amplitudes of the constrained inversion model are well controlled immediately after the first resistivity drop, although they are generally larger than that of the synthetic model at other depths. Whereas the resistivity texture aligns with the seismic section, the final MT data misfit remains similar compared with conventional inversion. The inverted base model represents a smooth resistivity distribution, which is similar to the output of conventional inversion except for the ripple-like texture due to the existence of the detail model. The detail model can be viewed as a spatial perturbation that introduces resistivity texture patterns.

Further discussion on the method

In the following, we further evaluate the constrained inversion from several perspectives. We demonstrate the necessity of the proposed algorithm configuration, examine the effect of inaccuracies in the seismic section on reconstruction, discuss the impact of hyperparameter adjustments on the inversion, and integrate the proposed method with model-based inversion, which simultaneously incorporates coarse-scale knowledge of geophysical body positions and shapes, as well as finer-scale knowledge of texture patterns in geophysical attribute distribution.

Algorithm settings

We first assess the necessity of the algorithm configuration described previously, specifically investigating whether the same inversion effect can be achieved with a simpler setup. Two potential simplifications are considered. First, the texture misfit penalty is excluded, and the seismic section is incorporated solely through a proper initial model. Second, only the detail model is used for parameterization, rather than a base and a detail model. Compared with the proposed objective function in equation 16, the objective functions for these potential alternatives are

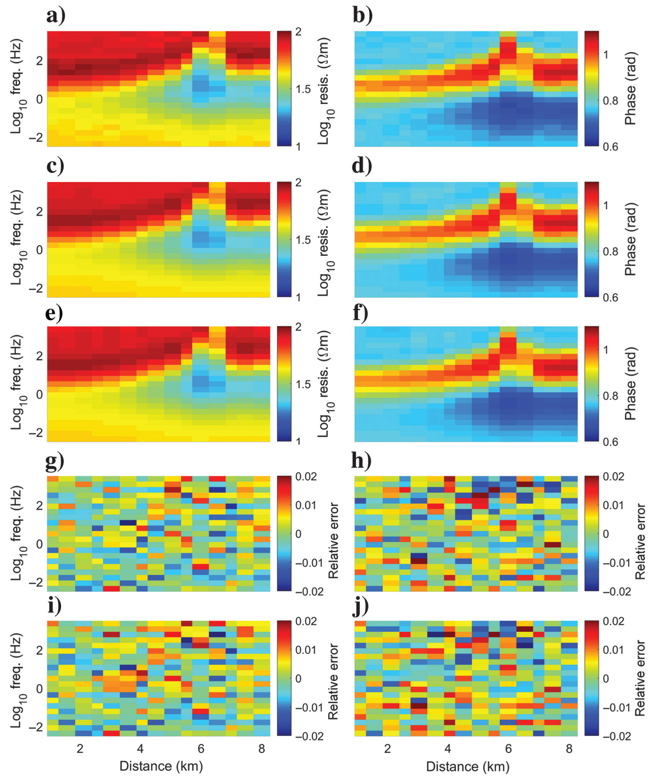

Figure 10. (a) Simulated apparent resistivity, (b) simulated impedance phase, (c) reconstructed apparent resistivity (conventional), (d) reconstructed impedance phase (conventional), (e) reconstructed apparent resistivity (constrained), (f) reconstructed impedance phase (constrained), (g) pixel-wise relative error between (a and c), (h) pixel-wise relative error between (b and d), (i) pixel-wise relative error between (a and e), and (j) pixel-wise relative error between (b and f).

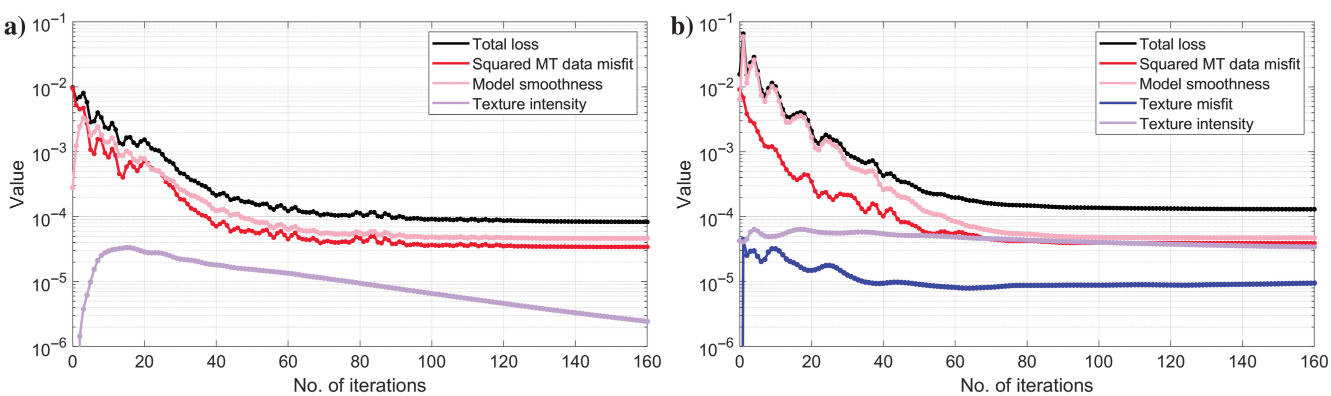

Figure 9. Convergence curves of the inversion with the (a) conventional method and (b) constrained method.

$$\mathcal{L}(\mathbf{m}_1, \mathbf{m}_2) = \alpha_D \|\mathbf{d}_{\text{obs}} - \mathcal{F}(\mathbf{m}_1 + \mathcal{I}_{c_2}(\mathbf{m}_2))\|^2 + \lambda_L \|\nabla(\mathbf{m}_1 + \mathcal{I}_{c_2}(\mathbf{m}_2))\|^2 + \lambda_M \|\mathbf{m}_2\|^2 \quad (20)$$

and

$$\mathcal{L}(\mathbf{m}_2) = \alpha_D \|\mathbf{d}_{\text{obs}} - \mathcal{F}(\mathcal{I}_{c_2}(\mathbf{m}_2))\|^2 + \lambda_L \|\nabla(\mathcal{I}_{c_2}(\mathbf{m}_2))\|^2 + \lambda_S \|\mathcal{T}(\mathbf{m}_{\text{seis}}) - \mathcal{T}(\mathbf{m}_2)\|^2 + \lambda_M \|\mathbf{m}_2\|^2. \quad (21)$$

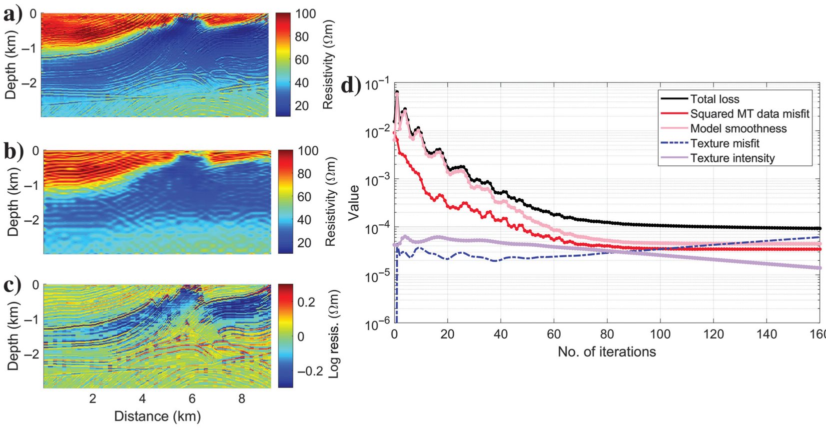

Figure 11. (a) Inverted model with the seismic texture constraint excluded, (b) base part of the inverted model, (c) detail part of the inverted model, and (d) convergence curve of the objective function.

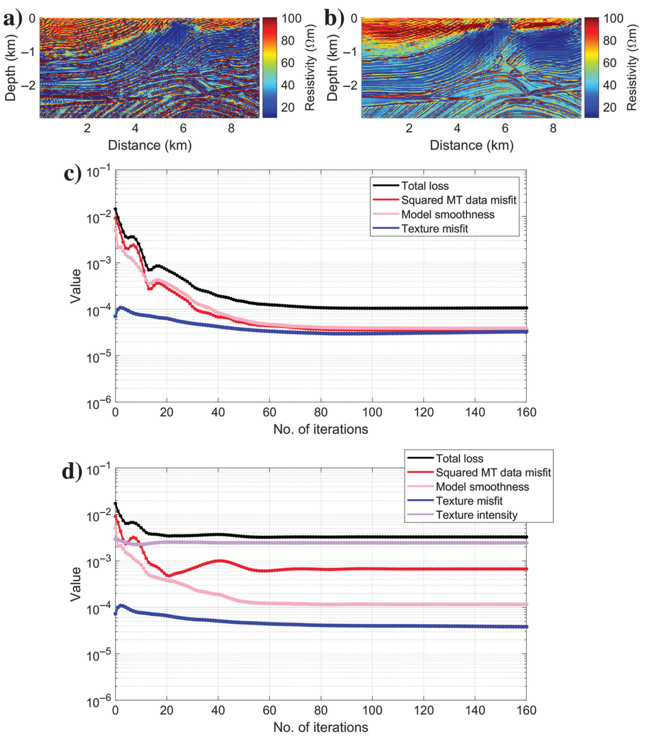

Figure 12. (a) Inverted model with only detail model parameterization $\lambda_M = 0$, (b) inverted model with only detail model parameterization $\lambda_M > 0$, (c) convergence curve in $\lambda_M = 0$ case, and (d) convergence curve in $\lambda_M > 0$ case.

We conduct numerical experiments to evaluate these two simplified implementations. In the first experiment, we use the same experiment setup as in the baseline inversion except that λ_S is set to zero. The inverted total model, base model, detail model, and convergence curves are shown in Figure 11. Compared with the proposed algorithm, the lack of texture constraint causes the detailed textures in the initial model to smear out in the inversion. Ultimately, the inversion output exhibits a smooth mosaic-like structure similar to conventional inversion (e.g., the low-resistivity bodies between 0.5 and 1.5 km in depth and at offsets between 5 and 8 km show little fine texture, as shown in Figure 11a). This is because the texture constraint introduces different update values for all the pixels in the detail model, without which the update gradient for the detail model is derived solely by multiplying the update gradient of the base model with a mapping matrix. This can also be validated in the convergence curve of the objective function in Figure 11d: the dashed blue line shows the weighted texture residual calculated with the same λ_S as in the baseline inversion (although not included in the objective function). Due to the lack of constraints, the dashed blue line shows a clear upward trend compared with the texture residual in Figure 9b. The final data misfit here is 0.59%.

In the second implementation, we set the texture intensity hyperparameter $\lambda_M = 0$ and $\lambda_M > 0$, respectively. The resulting inverted total models (i.e., detail models) and convergence curves are shown in Figure 12a–12d. The final data misfits are 0.59% and 2.2%, respectively. The results show that although the inversion with $\lambda_M = 0$ successfully fits the MT data, due to the absence of texture strength constraint, undesired and overly strong texture is presented. When the texture strength constraint is added, the inversion cannot converge satisfactorily because minimizing the $||\mathbf{m}_2||^2$ term seems to conflict with achieving the best MT data fit.

Discussion on the accuracy of seismic section

In practical scenarios, the quality of the seismic section can vary significantly due to factors such as measurement errors and inaccuracies in time-depth conversion. Here, we assume an extreme case where the correct seismic section is flipped horizontally or vertically (though such severe degradation is rare in practice). The initial and inverted models on fine and coarse grids, such as the base and detail models, are shown in Figure 13. The final data misfits for these two cases are 0.60% and 0.59%, respectively, with convergence curves shown in Figure 14. Despite the erroneous seismic sections and the incorporated irrelevant geophysical texture, the proposed inversions converge effectively, achieving final data misfits similar to those obtained without the section flips. This robustness arises from the fact that although the detail model affects the texture data, it has minimal impact on the MT data because the mapping from the fine grid to the coarse grid erases most of the fine seismic structure.

In the preceding experiments, texture errors are clearly noticeable in the final models. However, in practical scenarios, it can be challenging to identify subtle errors in the seismic section. Therefore, a high-precision seismic section is essential for the proposed method to function effectively.